

Sustentra Demo – Master Gap Descriptions

NY Part 253 Scope 1 Stationary Combustion – Food & Beverage Manufacturing

*Reference: Food_Manufacturing_Auditor_Checklist_Nora0605 | Regulatory Basis: NY Part 253;
40 CFR Part 98 Subpart C*

Criterion 1: Gaps cover all assertions (except presentation), especially complex or time-consuming manual processes

Completeness (2) — Gap 2 (pilot lights omitted, federal exemption cited), Gap 8 (billing evidence covers only partial year).

Accuracy (4) — Gap 3 (October volume 10x inflated), Gap 4 (boilers and diesel generator aggregated across different source types), Gap 5 (wrong EF applied to solid biomass months), Gap 10 (GWP-100 applied where GWP-20 required)

Occurrence (2) — Gap 6 (monthly biomass composite without weekly samples), Gap 7 (missing data filled without §3.1 substitution method).

Cutoff (1) — Gap 9 (cross-year billing period allocated entirely to current reporting year).

On complexity and time-consuming manual processes specifically: Gap 2 (pilot lights require knowing the NY Part 253 override of a federal exemption that explicitly permits omission — a multi-framework regulatory check), Gap 5 (biomass EF mismatch requires cross-referencing declared fuel type against the emission factor applied across twelve months of calculation inputs), Gap 6 (sampling chain verification requires tracing a biomass fraction result back through composite preparation records to weekly sample logs — each layer is a separate document check), Gap 10 (GWP basis verification requires knowing the NY Part 253 CO₂ Equivalent definition, its §496.5 cross-reference, and the specific GWP-20 values that differ from the federal default — none of this is visible without regulatory source knowledge). All four are genuinely time-consuming manual checks that an analyst would spend significant time on. This criterion is met.

Criterion 2: Evidence generated aligns with real-life examples

Gap 2 (pilot lights) — A company building its workbook from a federal template would find 40 CFR §98.30(d) and cite it as the basis for omission. This is a realistic and common failure mode for any facility that starts from EPA guidance without reading the state-specific override. The workbook footnote citing the federal exemption is exactly what a real company would write.

Gap 3 (October volume inflated) — Utility bill versus workbook transcription discrepancy. Transcription errors on high-volume utility data are extremely common in practice, particularly when companies manually transfer billing data into calculation workbooks without automated extraction.

Gap 4 (aggregation of boilers and generator) — Internal aggregation worksheets that group all combustion sources under a single label are standard practice at companies that prioritize

simplicity over regulatory compliance. The scenario of a company combining boilers and an emergency generator into one line item reflects how most internal GHG reporting templates are structured.

Gap 5 (wrong EF on biomass months) — Fuel delivery records clearly showing a seasonal fuel type switch while the workbook applies a uniform annual emission factor is a realistic and well-documented failure mode. Companies that switch fuels mid-year frequently fail to update their calculation methods correspondingly.

Gap 6 (single end-of-period lab report) — Submitting one result at the end of the sampling period with no underlying weekly logs is extremely common. Most companies treat the lab result as the deliverable and do not retain or submit the sampling chain behind it.

Gap 7 (missing data without §3.1 method) — A December bill not received before year-end deadline, filled with a simple monthly average, is one of the most common year-end reporting failures in practice. The §3.1 method is rarely known outside specialist practitioners.

Gap 8 (three of twelve billing months present) — A company submitting whatever bills were on hand at the time of compilation without recognizing that annual traceability is a specific regulatory requirement is realistic. This is particularly common at facilities where GHG reporting is handled by finance teams rather than environmental specialists.

Gap 9 (cross-year billing period not prorated) — Natural gas utility billing periods that span December and January are standard practice across major US utilities. Companies that use the billed figure directly without checking service period dates against the reporting year boundary make this error routinely.

Gap 10 (GWP-100 used) — A workbook footer citing 40 CFR Part 98 as the GWP basis is the default output of virtually every commercially available GHG calculation template. The NY Part 253 GWP-20 override is jurisdiction-specific and unknown to anyone who has not read the regulation directly. This is one of the most realistic scenarios in the set.

All ten gaps have evidence scenarios grounded in actual auditor and reporting experience. This criterion is met, with Gaps 7, 9, and 10 being particularly strong on real-world alignment.

Criterion 3: Gaps inserted correctly, not just low-tier "data missing" flags

Gap 8 (billing evidence covers only partial year) is the one gap with structural exposure on this criterion. Three of twelve billing months present reads as a missing documents flag at face value. The substantive reframe is that §253-2.7(l) imposes a specific traceability requirement — annual MMBtu must be demonstrably linked to billing statements by account — which the evidence pack fails. This is a regulatory compliance failure, not a generic absence flag, but the demo presentation must lead with the §253-2.7(l) framing for it to land that way. It is the weakest gap on this criterion and worth watching in the demo script.

Every other gap involves one of the following: a calculation error detectable by field-level rule matching (Gaps 3, 5, 10), an aggregation rule violation detectable by unit type comparison (Gap 4), a sampling chain break detectable by evidence document count and date analysis (Gap 6), a procedural compliance failure detectable by substitution method identification (Gap 7), a period allocation error detectable by billing date range extraction (Gap 9), or a jurisdiction-specific source omission with a self-incriminating regulatory citation detectable by regulatory reference extraction (Gap 2). None of these are simple absence flags. This criterion is met with Gap 8 as the single exception to monitor.

Criterion 4: Overall value prop for auditors — time-saving, complexity, practical usefulness

Time-saving: The highest time-saving gaps are Gap 6 (tracing a sampling chain across weekly logs, composite preparation records, and a lab report is among the most manual and time-consuming evidence checks in a GHG engagement), Gap 5 (checking emission factor consistency against declared fuel type across twelve monthly records requires reading every row of a fuel log and cross-referencing the calculation for each), Gap 2 (confirming pilot light treatment requires reading both the federal exemption and the state override and checking whether the company's citation is to the right jurisdiction), and Gap 10 (verifying the GWP basis requires knowing a NY-specific regulatory definition and its §496.5 cross-reference without any prompt from the client). These are all tasks an experienced analyst would spend significant time on and which the product surfaces immediately on upload.

Complexity: Gap 10 and Gap 2 are the most technically complex in the set — both require simultaneous knowledge of the federal framework and the NY Part 253 override, and both are invisible to any analyst working from generic GHG guidance. Gap 4 (aggregation unit type rules) and Gap 6 (sampling chain requirements) are also non-trivial and require regulatory source knowledge that goes beyond standard GHG Protocol training. The set as a whole demonstrates that the product handles regulatory reasoning, not just pattern matching or arithmetic checking.

Practical usefulness: The gaps collectively span the full audit workflow sequence. Gap 2 fires at source boundary review. Gaps 3 and 9 fire at extraction and reconciliation. Gap 4 fires at aggregation structure review. Gap 5 fires at emission factor validation. Gap 6 fires at evidence chain verification. Gap 7 fires at missing data procedure review. Gap 8 fires at evidence coverage completeness. Gap 10 fires at CO2e calculation methodology review. An auditor watching the demo sees the product adding value at every stage of their actual engagement workflow — not concentrated in one area and absent from others. This criterion is met.

Assertion Category Legend

Completeness	Accuracy	Occurrence	Presentation	Cutoff
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NEGLECT THIS Gap 1 Governing Regulatory Framework Incorrectly Disclosed; NY Part 253 Omitted from Methodology Statement [Presentation]

Checklist Workflow Area	Fuel Supplier and Billing Support / Boundary and Aggregation (Rows 10–11) — 253-2.7 introductory paragraph; 253-2.7(l); 253-1.3
Checklist Item	<i>The methodology statement or cover documentation must accurately describe the regulatory framework governing the emissions data report. For a facility reporting under NY Part 253, the governing standard is 6 NYCRR Part 253, which incorporates 40 CFR Part 98 Subpart C except as otherwise provided. A disclosure that cites only the federal framework without identifying the state governing regulation is incomplete and misleading to any reader relying on that disclosure to understand what requirements were applied.</i>
Gap Location	Workbook
In Workbook	Methodology tab or cover page — the workbook contains a written methodology statement that reads: "GHG emissions calculated in accordance with 40 CFR Part 98 Subpart C, General Stationary Fuel Combustion Sources, U.S. Environmental Protection Agency." No reference to 6 NYCRR Part 253, the New York State Department of Environmental Conservation, or any NY-specific requirements or overrides. The disclosure is technically accurate as a description of the calculation engine used but materially incomplete as a description of the governing regulatory framework for this submission.
In Evidence Pack	N/A
How It Is Embedded	NY Part 253 §253-2.7 states verbatim: "The operator of a facility who is required to report under section 1.2 of this Part must comply with subpart C of 40 CFR part 98 in reporting stationary fuel combustion emissions and related data to the department, except as otherwise provided in this section." The governing obligation is Part 253. The federal subpart is the incorporated calculation method, not the governing regulation. A methodology statement that identifies only 40 CFR Part 98 as the basis for reporting fails to disclose that NY Part 253 governs, that NY-specific overrides exist, and that the submission is directed to the DEC rather than the EPA. An auditor reading the methodology statement cannot determine from it whether any state-level requirements were identified or applied. This is a presentation failure because the reported information is not described in a way that allows an informed reader to understand the regulatory basis on which it was prepared. Critically, the incorrect disclosure in

	the methodology statement is also the root cause explanation for two other gaps in the demo — the GWP-100 error (Gap 11) and the pilot light omission — both of which flow directly from treating this as a federal filing rather than a state one. The methodology statement is self-incriminating: the system reads it, identifies the missing Part 253 reference, and flags it before any calculation review begins.
Auditor Relevance	Core to every NY Part 253 engagement. An auditor's first step on any engagement is confirming that the reporting entity understood and applied the correct governing standard.
Gap 2 Pilot Light Emissions Omitted; Federal Exemption Cited Where NY Part 253 Override Applies [<i>Completeness</i>]	
Checklist Workflow Area	Source Coverage (Row 1) – 253-2.7 core; 253-2.7(j); pilot lights row in the checklist
Checklist Item	<i>The operator must include emissions from pilot lights in the emissions data report. Engineering methods or common-fuel-source aggregation may be used when pilot lights are unmetered. Pilot lights are not exempt from reporting under NY Part 253 regardless of federal treatment</i>
Gap Location	Workbook + Evidence Pack
In Workbook	Equipment Inventory tab — no pilot light entries exist for any of the boilers or process heaters. A comment or methodology note in the workbook cites 40 CFR §98.30(d) as the basis for omission, which explicitly exempts pilot lights at the federal level. No pilot light row appears in the Emissions Summary.
In Evidence Pack	No engineering estimate for pilot light fuel consumption. No common-fuel-source aggregation note. No record of any attempt to quantify pilot light emissions. The absence of any documentation is consistent with a company that believed the federal exemption applied and made no further inquiry.
How It Is Embedded	40 CFR §98.30(d) states verbatim that reporting of GHG emissions from pilot lights is not required. NY Part 253 §253-2.7(j) overrides this directly: "The operator must include emissions from pilot lights in the emissions data report." The company built its workbook referencing the federal framework and cited the federal exemption without reading the state-specific override. The gap requires the system to simultaneously know the federal rule, know the NY override, and flag the citation in the workbook as an incorrect regulatory basis. No other

	gap in the set demonstrates this jurisdiction-switching capability as cleanly.
Auditor Relevance	Core to every NY Part 253 engagement. The federal exemption is real and widely known, making this a realistic and common failure mode for any company that prepares its report using federal guidance as the starting point.

Gap 3 October Fuel Volume 10x Inflated [Accuracy]	
Checklist Workflow Area	Individual Fuel Volume Reporting (Row 2) — 253-2.7(h)
Checklist Item	<i>Confirm individual fuel volumes are reported for combustion processes. Volumes must be traceable to billing or meter evidence and reconcile to the annual total.</i>
Gap Location	Workbook + Evidence Pack
In Workbook	Natural Gas tab — October natural gas volume is entered as a value 10x higher than the actual billed quantity (data entry transposition error). The annual total in the Emissions Summary reflects this inflated figure.
In Evidence Pack	October utility bill shows the correct lower volume. The discrepancy between the bill and the workbook entry is detectable at reconciliation.
How It Is Embedded	A transcription error inflated the October natural gas volume by a factor of ten. The billed quantity and the workbook entry do not reconcile. Sustentra surfaces this at the extraction and reconciliation stage by summing monthly bill values and comparing to the workbook annual input, flagging the month where the discrepancy originates.
Auditor Relevance	Core — data reconciliation to source documents is a fundamental verification check under both Part 253 and ISSA 5000.

Gap 4 Boilers and Diesel Generator Aggregated Across Different Source Types [Accuracy]	
Checklist Workflow Area	Boundary and Aggregation (Row 11) — 253-2.7(i); 253-2.7(m); 253-2.7(p)
Checklist Item	<i>Confirm units are aggregated only when allowed. Aggregation of stationary combustion units is limited to units of the same type. Different source categories cannot be grouped into a single reported line.</i>
Gap Location	Workbook + Evidence Pack
In Workbook	Emissions Summary tab — natural gas boilers (BLR-001, BLR-002, BLR-003) and the diesel emergency generator (GEN-001) are aggregated into a single combined CO2e row. These are different unit types and cannot be aggregated under Part 253.
In Evidence Pack	An aggregation worksheet groups both source categories together, providing the paper trail showing the company treated them as one group.
How It Is Embedded	NY Part 253 §253-2.7(i), incorporating 40 CFR §98.36(c), restricts aggregation of stationary combustion units to units of the same type. Boilers and emergency generators are different source categories. The workbook combines them into one line item in the Emissions Summary, which means the auditor cannot separately verify each source type's contribution or confirm that the correct calculation method was applied to each. Sustentra flags this when it reads the aggregation structure against the unit type declarations in the equipment inventory.
Auditor Relevance	Core — aggregation rules are an explicit Part 253 requirement and a direct constraint on how the Emissions Summary can be structured.

Gap 5 Wrong Emission Factor Applied to Solid Biomass Months [Accuracy]	
Checklist Workflow Area	Steam-Producing Units (Row 5, Conditional) — 253-2.7(a); Biomass / Biogas Calculation (Row 7, Conditional) — 253-2.7(e)

Checklist Item	<i>If biomass fuels are combusted, confirm the correct equation or tier is applied and that biogenic CO2 is calculated separately from fossil CO2. Natural gas emission factors cannot be applied to solid biomass combustion months.</i>
Gap Location	Workbook
In Workbook	Boiler tab — August and September rows reflect periods when the boiler co-fired solid biomass residues. Both months use the natural gas CO2 emission factor (53.06 kg CO2/MMBtu) rather than the biomass-specific method. The emissions for those months are materially understated and the biogenic/fossil split is absent.
In Evidence Pack	Fuel delivery records show solid biomass deliveries in August and September, providing the trigger evidence for the auditor.
How It Is Embedded	The boiler co-fired solid biomass residues during August and September. Part 253, incorporating 40 CFR Part 98 §98.33(e), requires separate calculation and reporting of biogenic CO2 from combined biomass and fossil fuel combustion, using the applicable biomass method rather than the natural gas emission factor. The workbook applied the natural gas EF for all twelve months uniformly, understating emissions for the biomass months and omitting the required biogenic/fossil split entirely. Sustentra detects the emission factor mismatch against the declared fuel type for those months.
Auditor Relevance	Core — biogenic vs. fossil separation for mixed combustion is a required calculation distinction under Part 253.

Gap 6 Monthly Biomass Composite Used Without Required Weekly Samples

[Occurrence]

Checklist Workflow Area	Biogenic Fuel Sampling (Row 8, Conditional) — 253-2.7(e)
Checklist Item	<i>For partially biogenic fuels analyzed monthly, confirm weekly samples were taken and combined into a homogenized, well-mixed monthly composite. A single end-of-period sample does not satisfy the requirement.</i>

Gap Location	Evidence Pack
In Workbook	No gap in the workbook itself — the biomass fraction value appears as a single figure. The problem is entirely in the evidence supporting that figure.
In Evidence Pack	One end-of-period lab report exists for the biomass fraction. No weekly sampling logs, no composite preparation record, no protocol documentation. The monthly result cannot be validated because the required underlying samples do not exist.
How It Is Embedded	Part 253 §253-2.7(e) requires that for partially biogenic fuels, weekly samples be taken throughout the month and combined into a documented monthly composite before the biomass fraction is applied. The evidence pack contains only a single lab result taken at the end of the measurement period, with no weekly samples behind it. The biomass fraction in the workbook therefore has no verifiable basis. Sustentra flags this at the occurrence assertion check when it finds a biomass fraction value with no supporting sampling chain.
Auditor Relevance	Core when biomass combustion is present — sampling frequency for biogenic fuels is explicitly required under Part 253.

Gap 7 Missing Data Filled Without Part 253 §3.1 Substitution Method [Occurrence]	
Checklist Workflow Area	Missing Data (Row 12) — 253-2.7(n); 253-3.1
Checklist Item	<i>If emissions data are missing, confirm substitution follows section 3.1 rather than ad hoc assumptions. The substitution method and rationale must be documented.</i>
Gap Location	Workbook + Evidence Pack
In Workbook	Natural Gas tab — December natural gas volume is filled with a simple average of the prior eleven months. A workbook note reads 'estimated — bill not received.' No reference to Part 253 §3.1 and no documentation of the substitution basis.

In Evidence Pack	No December utility bill. No missing data log. No substitution rationale document. The absence of both the bill and the required documentation is the gap.
How It Is Embedded	The December natural gas bill was not received before the reporting deadline. The company filled the gap using a simple monthly average, which is an undocumented ad hoc method. Part 253 §3.1 specifies the required missing data substitution procedures, which must be followed and documented. A simple average applied without reference to the prescribed method is a nonconformance regardless of whether the estimated value is numerically close to the actual figure. Sustentra flags this at the occurrence check when it detects a substituted value with no §3.1 method reference or supporting documentation.
Auditor Relevance	Core — Part 253 §3.1 missing data procedures are a mandatory requirement whenever a gap in calculation inputs exists.

Gap 8 Billing Evidence Covers Only Partial Year [<i>Completeness</i>]	
Checklist Workflow Area	Fuel Supplier and Billing Support (Row 10) — 253-2.7(I)
Checklist Item	<i>Confirm supplier name, customer account number, service account ID, annual MMBtu delivered to each account, and billing-statement support. Annual traceability by account is required.</i>
Gap Location	Evidence Pack
In Workbook	No gap in the workbook — the annual natural gas total is reported as a single figure. The problem is that the evidence supporting it is incomplete.
In Evidence Pack	Only three monthly utility bills are present in the evidence pack (January, February, March). Nine months of billing statements are absent. No annual account summary from the supplier. The annual MMBtu figure in the workbook cannot be traced to source documents for the majority of the reporting year.

How It Is Embedded	Part 253 §253-2.7(l) requires that the operator report annual MMBtu delivered to each account according to billing statements, with full supplier and account identification. The evidence pack supports only three of twelve months, leaving nine months of consumption without documentary basis. Unlike Gap 8 (which is about the method used to fill a known gap), this gap is about the absence of required evidence for most of the year with no acknowledged gap or substitution at all. Sustentra flags this at the completeness check when the number of billing documents does not cover the full reporting period.
Auditor Relevance	Core — annual billing evidence traceability is an explicit data requirement under Part 253 §253-2.7(l).

Gap 9 Cross-Year Billing Period Allocated Entirely to Current Reporting Year [Cutoff]	
Checklist Workflow Area	Individual Fuel Volume Reporting (Row 2) — 253-2.7(h); Boundary and Aggregation (Row 11) — 253-2.7(i)
Checklist Item	<i>Fuel volumes must be reported for the calendar year in which they were consumed. NY Part 253 defines the reporting period as January 1 through December 31. Cross-year billing periods must be prorated to the correct calendar year.</i>
Gap Location	Workbook + Evidence Pack
In Workbook	Natural Gas tab — the December row records the full volume from a billing statement dated December 16 through January 15 of the following year. No proration was applied. Approximately half a month of the following year's consumption is included in the current year's total, and the same volume will be double-counted in next year's report if the same approach is used.
In Evidence Pack	The December utility bill clearly shows a service period of December 16 through January 15, spanning two calendar years. This is the trigger document — the billing dates make the cross-year period visible.

How It Is Embedded	The December natural gas billing statement covers a period that crosses the December 31 / January 1 boundary. Part 253 requires reporting on a calendar year basis. The correct treatment is to prorate the billed volume between the two years based on the number of days each period falls within the reporting year. The company recorded the full billed volume in December without proration, overstating current-year consumption by approximately 16 days of usage and creating a double-count risk in the following year. Sustentra flags this at the cutoff check when it reads the billing statement service dates and detects a period extending beyond December 31.
Auditor Relevance	Core — period allocation to the correct calendar year is a fundamental requirement for any calendar-year reporting framework, and cross-year billing periods are common in utility billing.

Gap 10 GWP-100 Applied Where NY Part 253 Requires GWP-20 [Accuracy]	
Checklist Workflow Area	Individual Fuel Volume Reporting / Calculation Method (Row 2, Row 3) — 253-2.7(h); 253-2.7(c)
Checklist Item	<i>NY Part 253 defines CO2 equivalent as a 20-year global warming potential basis, pointing to the CO2e values in 6 NYCRR §496.5. The federal GHGRP uses GWP-100. These produce materially different values for CH4. A NY Part 253 report must use the §496.5 values, not the federal default.</i>
Gap Location	Workbook
In Workbook	Emissions Summary tab — the CO2e column applies GWP-100 values (CH4 = 28, N2O = 265), which are the EPA GHGRP / IPCC AR5 federal defaults. A note in the workbook footer cites 40 CFR Part 98 as the basis for the GWP values. The correct values under NY Part 253 via 6 NYCRR §496.5 are CH4 = 84, N2O = 264. CH4 under the correct NY basis is exactly 3x higher than the federal GWP-100 value. The error propagates materially to the reported CO2e total.
In Evidence Pack	No gap in the evidence pack — this is a calculation methodology error in the workbook

How It Is Embedded	NY Part 253 defines CO2 equivalent as "the amount of carbon dioxide by mass that would produce the same global warming impact as a given mass of another GHG over an integrated 20-year time frame after emission. CO2e shall be calculated using the CO2e values in section 496.5 of this Title." Section 496.5 of 6 NYCRR Part 496, published by DEC, specifies CH4 = 84 and N2O = 264 using IPCC AR5 GWP-20. The company built its workbook referencing the federal standard (GWP-100, CH4 = 28), which is the default in 40 CFR Part 98 and the more commonly known metric. Because CH4's GWP-20 value is exactly 3x its GWP-100 value, any facility with meaningful CH4 emissions will materially understate its CO2e. This is a NY-specific requirement invisible to anyone working from federal guidance alone, and the incorrect GWP basis is self-declared in the workbook footer, making it a clean document-based detection.
Auditor Relevance	Core to every NY Part 253 engagement. GWP basis is a defined term in the regulation with a direct cross-reference to §496.5. The specific values are now verified against the primary DEC source document. The CH4 discrepancy (28 vs. 84) is the headline number — N2O moves by only 1 point (265 to 264) and is immaterial for demo purposes.